1) $A = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \end{pmatrix}$

a) berechne <u>alle</u> x s.d. $\|Ax - b\|_2 \to \min$, $b = \begin{pmatrix} b_1 \\ b_2 \end{pmatrix}$

($\|Ax-b\|_2 \to \min \iff \underline{A^T A x = A^T b}$)

b) A^+ (alle Lsg x von a) und suche kürzestes x)

c) $\sigma_i(A) > 0$ (EW λ_i von $A^T A$ oder $A A^T$ $\Rightarrow \sigma_i = \sqrt{\lambda_i}$)

a) $A^T A = \begin{pmatrix} 0 & 1 \\ 1 & 0 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 2 \end{pmatrix}$

$A^T b = \begin{pmatrix} 0 & 1 \\ 1 & 0 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} b_1 \\ b_2 \end{pmatrix} = \begin{pmatrix} b_2 \\ b_1 \\ b_1 + b_2 \end{pmatrix}$

$$\left(\begin{array}{ccc|c} 1 & 0 & 1 & b_2 \\ 0 & 1 & 1 & b_1 \\ 1 & 1 & 2 & b_1 + b_2 \end{array}\right) \iff \left(\begin{array}{ccc|c} 1 & 0 & 1 & b_2 \\ 0 & 1 & 1 & b_1 \\ 0 & 1 & 1 & b_1 \end{array}\right)$$

$x_3 = \alpha, \quad \alpha \in \mathbb{R}$

$x_2 + x_3 = b_1 \Rightarrow x_2 = b_1 - \alpha$

$x_1 + x_3 = b_2 \Rightarrow x_1 = b_2 - \alpha$

$$x = \begin{pmatrix} b_2 - \alpha \\ b_1 - \alpha \\ \alpha \end{pmatrix} = \underline{\begin{pmatrix} b_2 \\ b_1 \\ 0 \end{pmatrix} - \alpha \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix}} \quad \alpha \in \mathbb{R}$$

b) $\|x\|_2 \to \min$

$$\Big\| \underbrace{\begin{pmatrix} b_2 \\ b_1 \\ 0 \end{pmatrix}}_{\hat{b}} - \alpha \underbrace{\begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix}}_{\hat{A}} \Big\|_2 \to \min$$

$\|\hat{A}\alpha - \hat{b}\|_2 \to \min$

$$\hat{A}^T \hat{A}\, \alpha = \hat{A}^T \hat{b}$$

$$(1,1,-1)\begin{pmatrix}1\\1\\-1\end{pmatrix}\alpha = (1,1,-1)\begin{pmatrix}b_2\\b_1\\0\end{pmatrix}$$

$$3\alpha = b_2 + b_1$$

$$\alpha = \frac{b_1+b_2}{3}$$

$$x = \begin{pmatrix}b_2\\b_1\\0\end{pmatrix} - \frac{b_1+b_2}{3}\begin{pmatrix}1\\1\\-1\end{pmatrix}$$

$$= \frac{1}{3}\begin{pmatrix}3b_2 - (b_1+b_2)\\3b_1 - (b_1+b_2)\\0 - (-1)(b_1+b_2)\end{pmatrix}$$

$$= \frac{1}{3}\begin{pmatrix}-b_1 + 2b_2\\2b_1 - b_2\\b_1 + b_2\end{pmatrix} = \underbrace{\frac{1}{3}\begin{pmatrix}-1 & 2\\2 & -1\\1 & 1\end{pmatrix}}_{A^+}\begin{pmatrix}b_1\\b_2\end{pmatrix}$$

c) $\sigma_i(A) > 0$: $\sigma_i^2 = \lambda_i$, λ_i EW von $\underline{A^TA} \rightarrow 3\times 3$

bzw EW von $\underline{AA^T} \rightarrow 2\times 2$

$$A^TA = \begin{pmatrix}1&0&1\\0&1&1\\1&1&2\end{pmatrix}$$

$A: 2\times 3$

$$0 = \det(A^TA - \lambda I) = \det\begin{pmatrix}1-\lambda & 0 & 1\\0 & 1-\lambda & 1\\1 & 1 & 2-\lambda\end{pmatrix}$$

$$= (1-\lambda)\det\begin{pmatrix}1-\lambda & 1\\1 & 2-\lambda\end{pmatrix} + \det\begin{pmatrix}0 & 1-\lambda\\1 & 1\end{pmatrix}$$

$$= (1-\lambda)\big((1-\lambda)(2-\lambda) - 1\big) + \big(0 - (1-\lambda)\big)$$

$$= (1-\lambda)\left(\cancel{2} - 3\lambda + \lambda^2 - \cancel{1} - \cancel{1}\right)$$

$$= (1-\lambda)(\lambda^2 - 3\lambda) = (1-\lambda)\,\lambda\,(\lambda-3)$$

$\lambda_1 = 3$, $\lambda_2 = 1$, $\lambda_3 = 0$

$\sigma_1 = \sqrt{3}$ $\quad$ $\sigma_2 = 1$

A^TA

$$A A^T = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \\ 1 & 1 \end{pmatrix} =$$

$$= \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$$

$$0 = \det \begin{pmatrix} 2-\lambda & 1 \\ 1 & 2-\lambda \end{pmatrix} = (\lambda - 2)^2 - 1$$

$$\lambda - 2 = \pm 1$$

$$\underline{\lambda_1 = 3} \qquad \underline{\lambda_2 = 1}$$

$$\underline{\underline{\sigma_1 = \sqrt{3}}} \qquad \underline{\sigma_2 = 1}$$